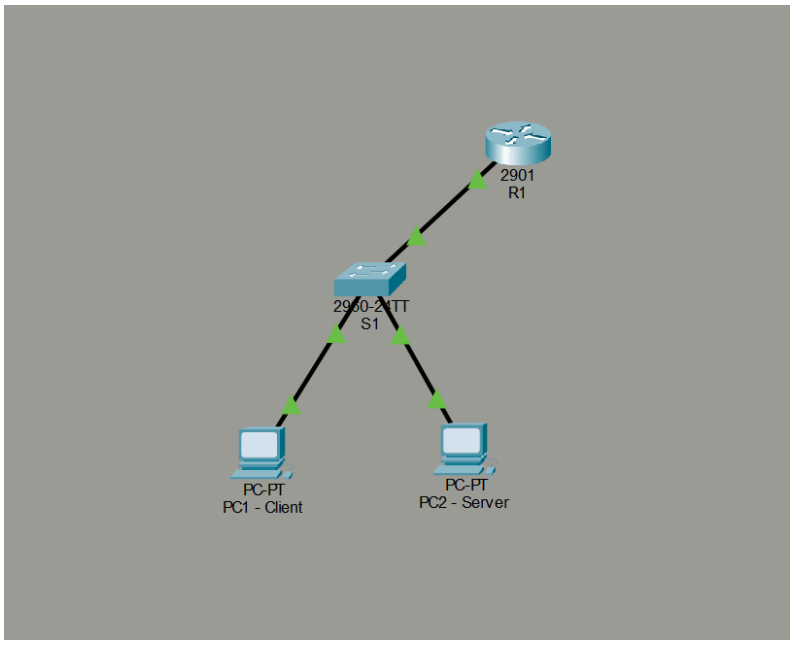
Lab3 Report — MIME File Transfer over Router-on-a-Stick with Wireshark Analysis

สมาชิกในกลุ่ม
พิสิษฐ์ ทรัพย์อุดมโชติ 673380285-2 sec.1



Configure packet tracer file:
https://drive.google.com/file/d/1m9hO4y7jpHS68bMb2Xj3wK_y9vbQfqFH/view?usp=sharing

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📘 Lab Overview

Scenario

A client in one VLAN transfers files to a server in another VLAN using a custom MIME-based protocol (JSON header + binary payload). You will verify connectivity, observe encapsulation, and analyze packets using Wireshark.

This lab reinforces how Layer 2 segmentation, Layer 3 routing, TCP reliability, and application metadata interact.

🗺️ Topology

```
None
PC1 (VLAN 10) ----\
                    \ access
                     \
                     [S1]---- trunk ----[R1]
                     /
                    / access
PC2 Server (VLAN 20)
```

📋 Addressing Scheme

Device	Interface	IP	Gateway
R1	G0/0.10	192.168.10.1/24	—
R1	G0/0.20	192.168.20.1/24	—
PC1-Client	NIC	192.168.10.10	192.168.10.1
PC2-Server	NIC	192.168.20.20	192.168.20.1

🧠 Background Theory

Router-on-a-Stick

A single physical interface carries multiple VLANs via 802.1Q tagging. Subinterfaces perform routing between broadcast domains.

MIME Metadata

MIME labels content type (e.g., image/png). Your protocol wraps files with JSON describing:

- MIME type
- Size

Why This Matters

Real systems — APIs, storage services, message brokers — depend on this exact pattern: structured metadata + binary payload over reliable transport.

🔧 Part 1 — Configure Network Devices

Step 1 — Router Configuration

```
None
enable
configure terminal
hostname R1

interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

end
write memory
```

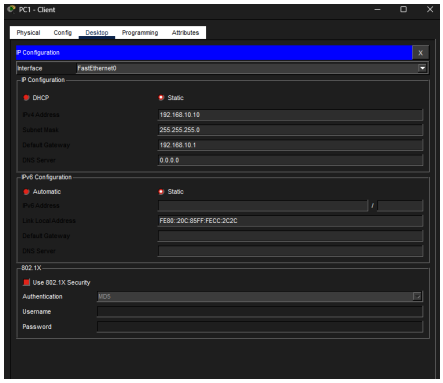
Step 2 — Switch Configuration

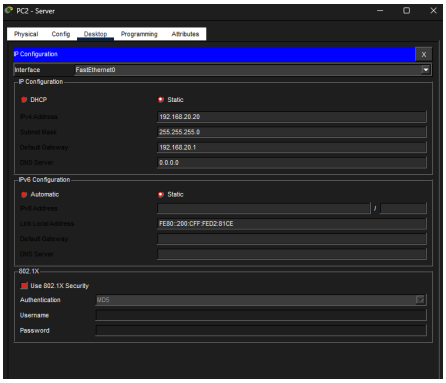
```
None
enable
configure terminal
hostname S1

vlan 10
name CLIENTS
```

```
vlan 20
name SERVERS

interface g0/1
switchport mode trunk

interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 20

end
write memory
```

Step 3 — Configure Hosts

Set static IP addresses per table.





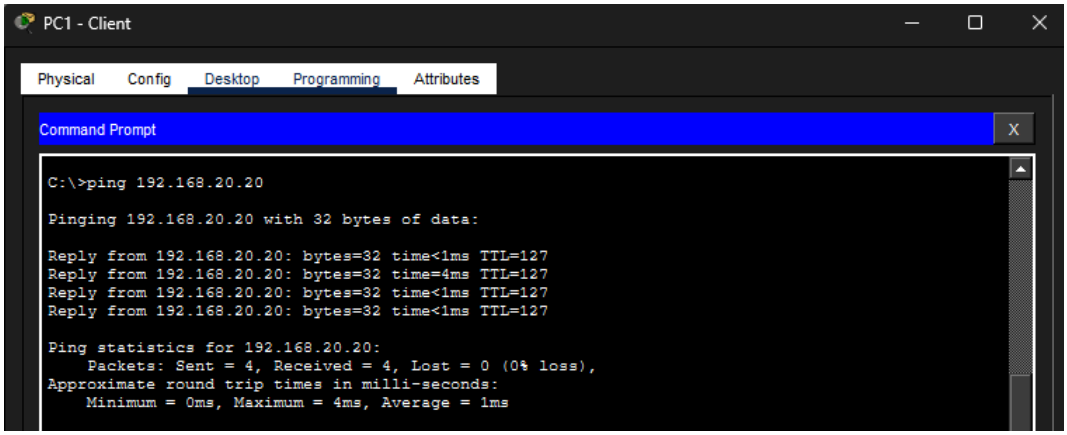
🔍 Part 2 — Verify Connectivity

From PC1:

```
None
ping 192.168.20.20
```

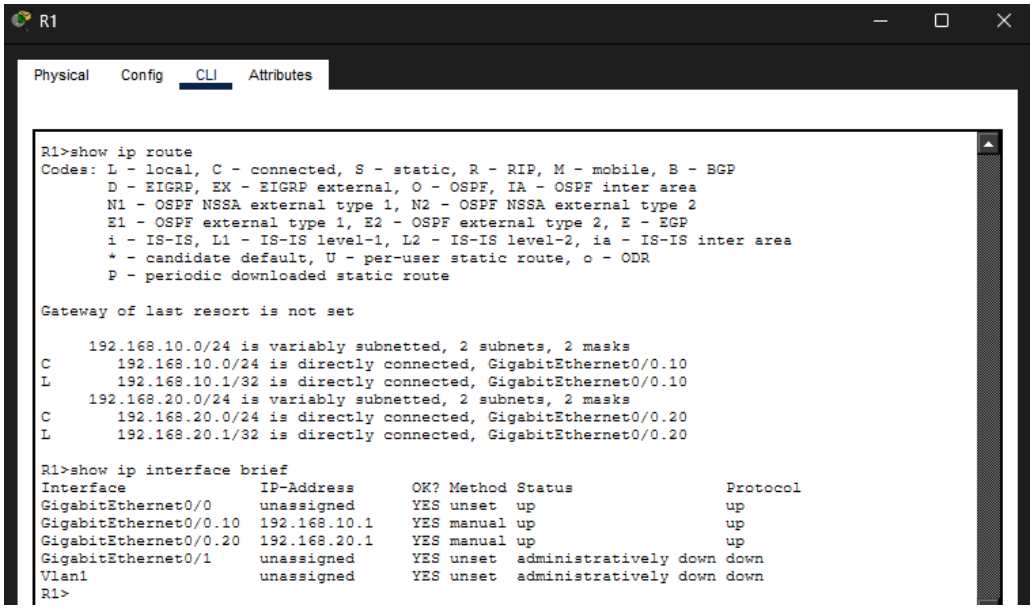
Verify:

- Successful replies
- ARP entries



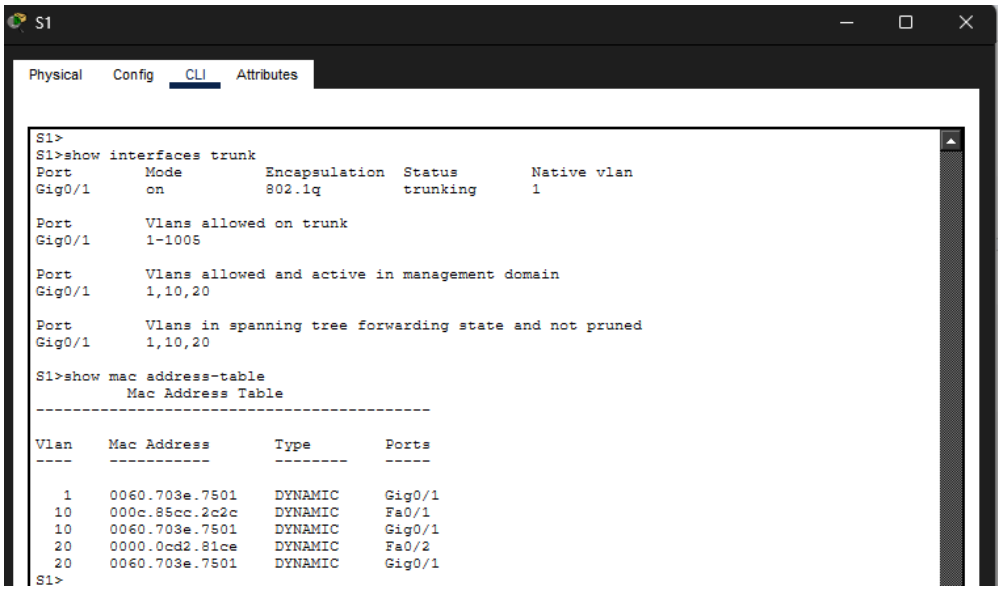
Router checks:

```
None
show ip route
show ip interface brief
```



Switch checks:

```
None
show interfaces trunk
show mac address-table
```



🧪 Part 3 — Deploy MIME Server

On Server (PC2)

```
None
cd automation
python server/main_enhanced.py --host 0.0.0.0 --port 65432 --verbose
```

On Client (PC1)

```
None
cd automation
python client/main_enhanced.py assets/sample.png --host 192.168.20.20 --port
65432
```

```
C:\Users\acer\OneDrive\Desktop\network\week01-mime-typing-main> python client/main_enhanced.py asset
s/sample.png --host 127.0.0.1 --port 65432
[21:40:19] INFO: Starting client with 1 files to 127.0.0.1:65432
[21:40:19] INFO: Connected to 127.0.0.1:65432
[21:40:19] INFO: Sent assets/sample.png (108 bytes) as image/png
[21:40:19] INFO: Connection closed. Sent: 1, Failed: 0
[21:40:19] INFO: All files sent successfully
```

Confirm file saved.

📡 Part 4 — Wireshark Capture Walkthrough on PC1 and PC1

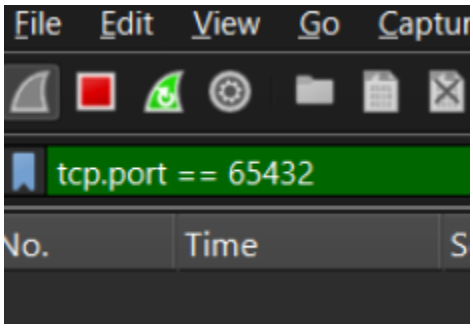
Step 1 — Start Capture

Open Wireshark.

Select interface connected to network.

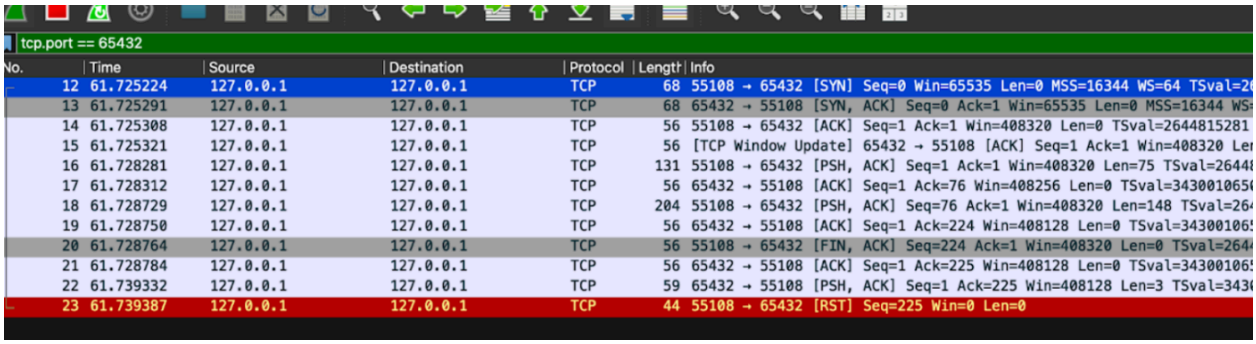
Apply capture filter:

```
None
tcp port 65432
```



Step 2 — Initiate Transfer

Send file again.



tcp.port == 65432

No.	Time	Source	Destination	Protocol	Length	Info
12	61.725224	127.0.0.1	127.0.0.1	TCP	68	55108 → 65432 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=2
13	61.725291	127.0.0.1	127.0.0.1	TCP	68	65432 → 55108 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS
14	61.725308	127.0.0.1	127.0.0.1	TCP	56	55108 → 65432 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=2644815281
15	61.725321	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 65432 → 55108 [ACK] Seq=1 Ack=1 Win=408320 Le
16	61.728281	127.0.0.1	127.0.0.1	TCP	131	55108 → 65432 [PSH, ACK] Seq=1 Ack=1 Win=408320 Len=75 TSval=2644
17	61.728312	127.0.0.1	127.0.0.1	TCP	56	65432 → 55108 [ACK] Seq=1 Ack=76 Win=408256 Len=0 TSval=3430010650
18	61.728729	127.0.0.1	127.0.0.1	TCP	204	55108 → 65432 [PSH, ACK] Seq=76 Ack=1 Win=408320 Len=148 TSval=264
19	61.728750	127.0.0.1	127.0.0.1	TCP	56	65432 → 55108 [ACK] Seq=1 Ack=224 Win=408128 Len=0 TSval=3430010650
20	61.728764	127.0.0.1	127.0.0.1	TCP	56	55108 → 65432 [FIN, ACK] Seq=224 Ack=1 Win=408320 Len=0 TSval=2644
21	61.728784	127.0.0.1	127.0.0.1	TCP	56	65432 → 55108 [ACK] Seq=1 Ack=225 Win=408128 Len=0 TSval=3430010650
22	61.739332	127.0.0.1	127.0.0.1	TCP	59	65432 → 55108 [PSH, ACK] Seq=1 Ack=225 Win=408128 Len=3 TSval=3430
23	61.739387	127.0.0.1	127.0.0.1	TCP	44	55108 → 65432 [RST] Seq=225 Win=0 Len=0

Stop capture after completion.

Step 3 — Analyze Packets

A — ARP Exchange

Look for:

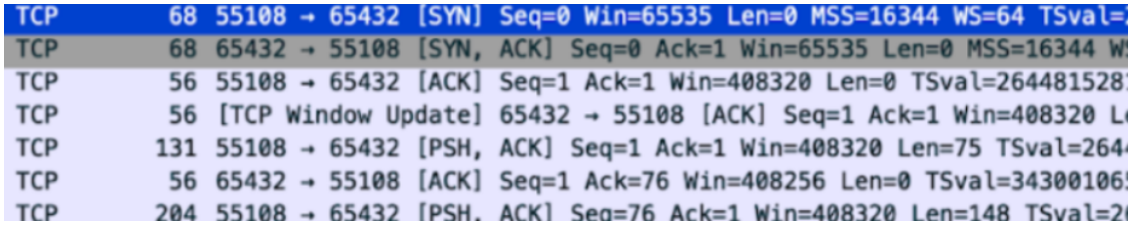
- Who has 192.168.20.20?
  เพราะเป็นใน loopback แทนการส่งข้าม VLAN จึงเป็นเครื่องตัวเอง 127.0.0.1

Confirm MAC resolution.

B — TCP Three-Way Handshake

Identify:

1 SYN 2 SYN-ACK 3 ACK



TCP	68	55108 → 65432 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=
TCP	68	65432 → 55108 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 W
TCP	56	55108 → 65432 [ACK] Seq=1 Ack=1 Win=408320 Len=0 TSval=264481528
TCP	56	[TCP Window Update] 65432 → 55108 [ACK] Seq=1 Ack=1 Win=408320 L
TCP	131	55108 → 65432 [PSH, ACK] Seq=1 Ack=1 Win=408320 Len=75 TSval=264
TCP	56	65432 → 55108 [ACK] Seq=1 Ack=76 Win=408256 Len=0 TSval=34300106
TCP	204	55108 → 65432 [PSH, ACK] Seq=76 Ack=1 Win=408320 Len=148 TSval=2

Check sequence numbers.

C — Data Transfer

Follow stream:

Right click → Follow → TCP Stream.

You should see JSON header similar to:

```
None
{"mime_type":"image/png","size":12345}
```

```
{"mime_type": "image/png", "size": 108}
.PNG
.
xxxx
```

🔬 Packet Dissection Guide

Layer	Field	What to Observe
Ethernet	Src/Dst MAC	Frame direction
802.1Q	VLAN ID	Segmentation
IP	TTL	Routing behavior
TCP	Flags	Reliability
Payload	JSON	MIME metadata

🛠️ Part 5 — Troubleshooting Labs

Scenario 1 — Broken Routing

Shutdown subinterface.

Observe:

```
C:\Users\acer\OneDrive\Desktop\network\week01-mime-typing-main> py client/main_enhanced.py assets/sample.png --host 127.0.0.1 --port 65432
[19:54:28] INFO: Starting client with 1 files to 127.0.0.1:65432
[19:54:30] ERROR: Failed to connect to 127.0.0.1:65432: [WinError 10061] No connection could be made because the target
machine actively refused it
[19:54:30] ERROR: Some files failed to send
```

- Ping failure
- TCP timeout

Scenario 2 — Firewall Block

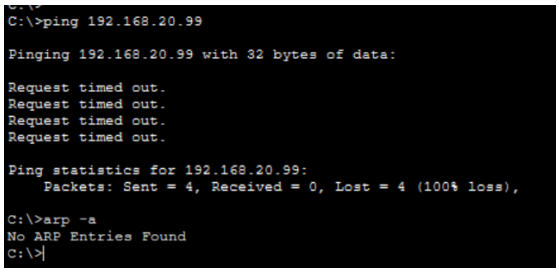
Block port.

Observe: เนื่องจากใช้ Localhost Firewall จึงไม่มีการบล็อก ทำให้การเชื่อมต่อสำเร็จ(bypass firewall) และไม่มีการส่งแพ็กเก็ต SYN ซ้ำ

- SYN retransmissions

Scenario 3 — Wrong Gateway

Observe:



```
C:\>ping 192.168.20.99

Pinging 192.168.20.99 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.99:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>arp -a
No ARP Entries Found
C:\>
```

- ARP incomplete

Scenario 4 — Packet Loss Simulation

Drop packets.

Observe: ทดสอบไม่ได้เพราะทำบน local host

- TCP retransmissions
- Window adjustment

📝 Lab Report Requirements

Include:

- Screenshots of configurations
- Ping results
- Wireshark capture screenshots
- JSON header evidence
- Encapsulation explanation
- Troubleshooting notes

🎓 Assessment Questions

1. Why does TCP retransmit lost segments?
   TCP ส่งข้อมูลแบบ reliable (เชื่อถือได้) ถ้า packet หาย จะต้องส่งใหม่เรื่อยๆ

2. How does VLAN tagging affect broadcast domains?
   VLAN tagging ทำให้ เครือข่ายเดียว แยกเป็นหลาย broadcast domain

3. What role does MIME play compared to TCP headers?
   MIME สนว่าข้อมูลคืออะไร แต่ TCPสนแค่ว่าทำยังไงจะส่งไปถึง

4. Where does routing decision occur?
   ทำการตรวจ destination IP แล้วเลือก interface ที่จะส่งออก

5. Why does ARP precede TCP communication?
   ต้องรู้ MAC ก่อน ถึงจะส่ง frame ได้ ARP จึงต้องทำงานเพื่อจับคู่ IP ให้เป็น MAC Address ให้สำเร็จก่อนที่การทำ TCP Handshake จะเริ่มขึ้น

📊 Reflection Exercise

Explain:

> How does application meaning survive multiple encapsulation layers?
> การส่งข้อมูลนี้ในแต่ละ layer จะไม่ได้ทำอะไรกับตัวข้อมูล เหมือนข้อมูลเป็นจดหมายแค่ห่อเพิ่มอีกชั้นพร้อมทำหน้าที่ของ layer นั้นๆให้เสร็จลุล่วงตามชั้นที่ห่อเพิ่มจนส่งถึงจุดหมาย

⚠️ Common Pitfalls

- Server bound to localhost only
- Missing trunk
- Wrong VLAN
- OS firewall enabled
- Capture on wrong interface

🚀 Extension Activities

- Add TLS encryption
- Measure throughput vs latency
- Capture checksum errors
- Add checksum field to protocol

- Containerize server
- Compare with HTTP transfer

🧬 Instructor Notes — What This Lab Really Teaches

This is not just networking.

It demonstrates:

- Protocol layering philosophy
- Reliability engineering
- Metadata semantics
- Distributed system fundamentals

The student watches a file become:

> intention → bytes → frames → signals → bytes → meaning

That transformation is the essence of computer networks.